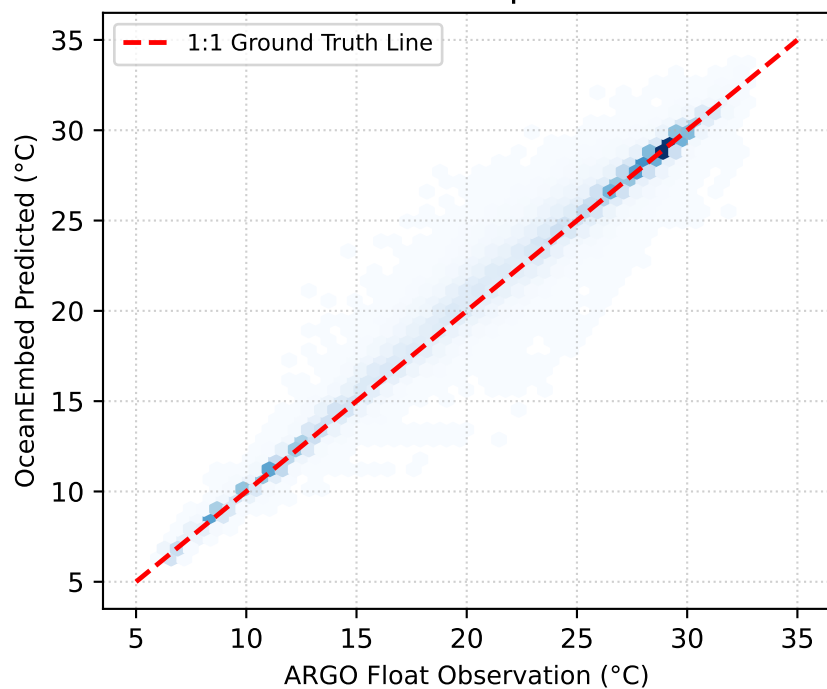


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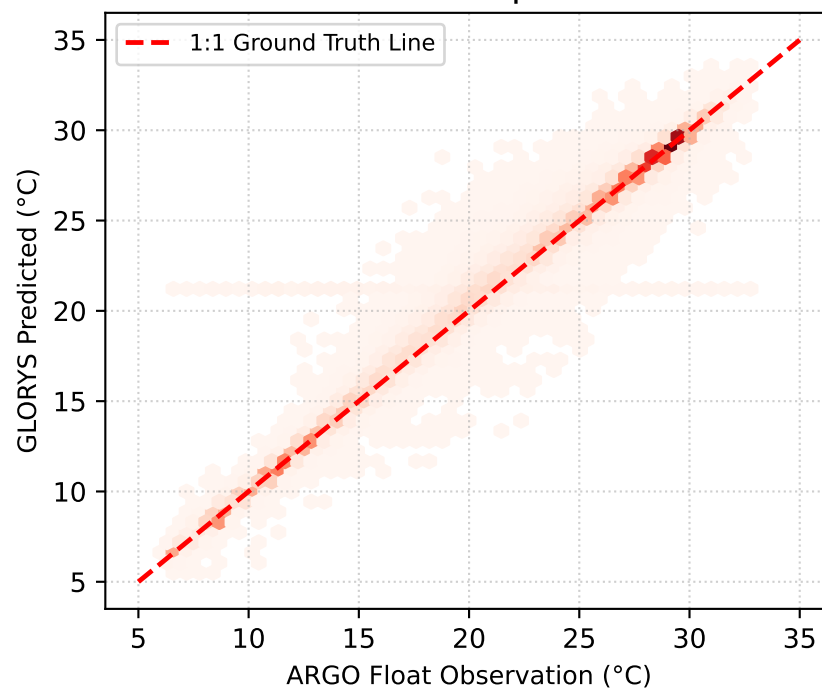
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| Metric                        | OceanEmbed (V2_8Ch) | Copernicus GLORYS12V1 | Delta / Margin |
|-------------------------------|---------------------|-----------------------|----------------|
| Root Mean Square Error (RMSE) | 0.9336 °C           | 1.2059 °C             | -0.2723 °C     |
| Mean Absolute Error (MAE)     | 0.5899 °C           | 0.6036 °C             | -0.0137 °C     |
| Mean Bias (Systematic Drift)  | +0.2345 °C          | +0.2471 °C            | -0.0125 °C     |
| Coefficient of Determ. (R²)   | 0.9827              | 0.9712                | +0.0115        |
| Pearson Correlation (r)       | 0.9919              | 0.9861                | +0.0058        |
| Max Observed Error            | 8.90 °C             | 14.95 °C              | -6.05 °C       |

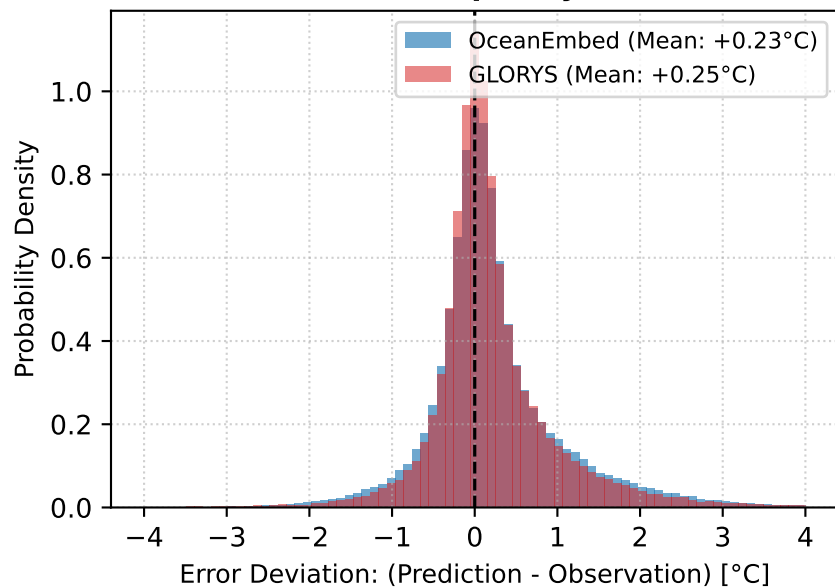
**OceanEmbed vs In-Situ ARGO**  
**RMSE: 0.9336°C | R<sup>2</sup>: 0.9827**



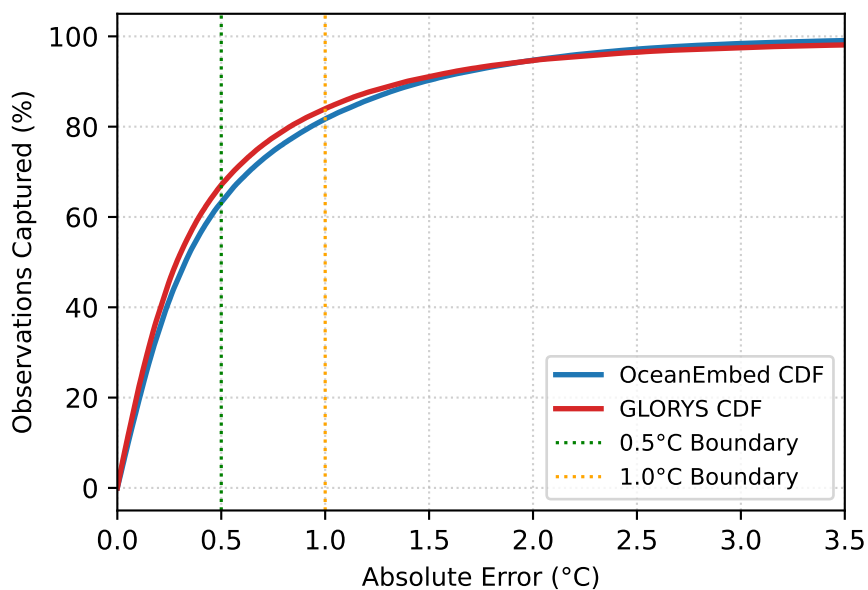
**GLORYS12V1 vs In-Situ ARGO**  
**RMSE: 1.2059°C | R<sup>2</sup>: 0.9712**

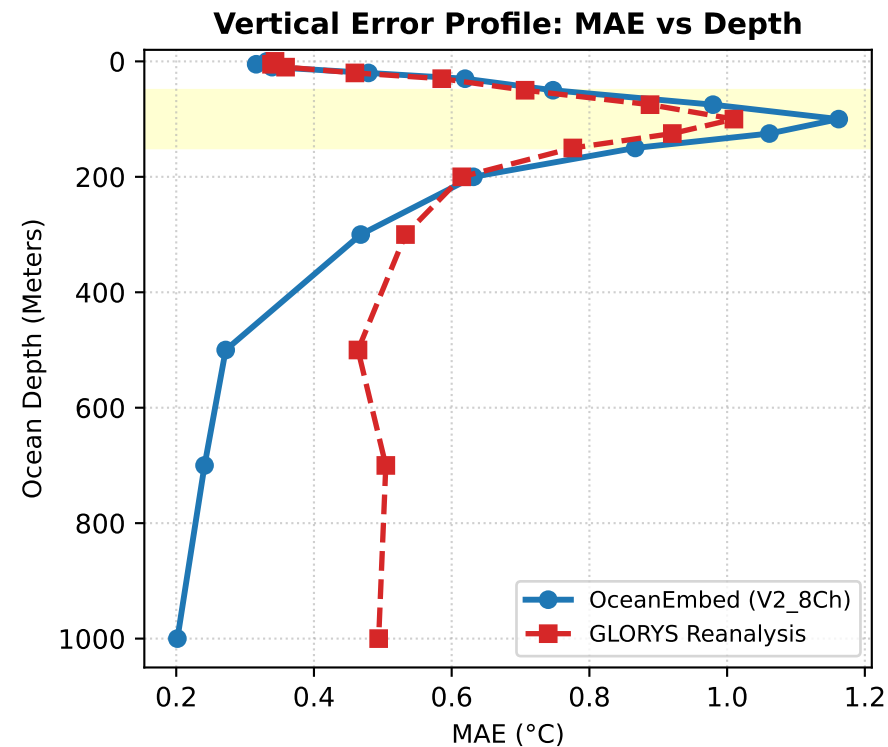
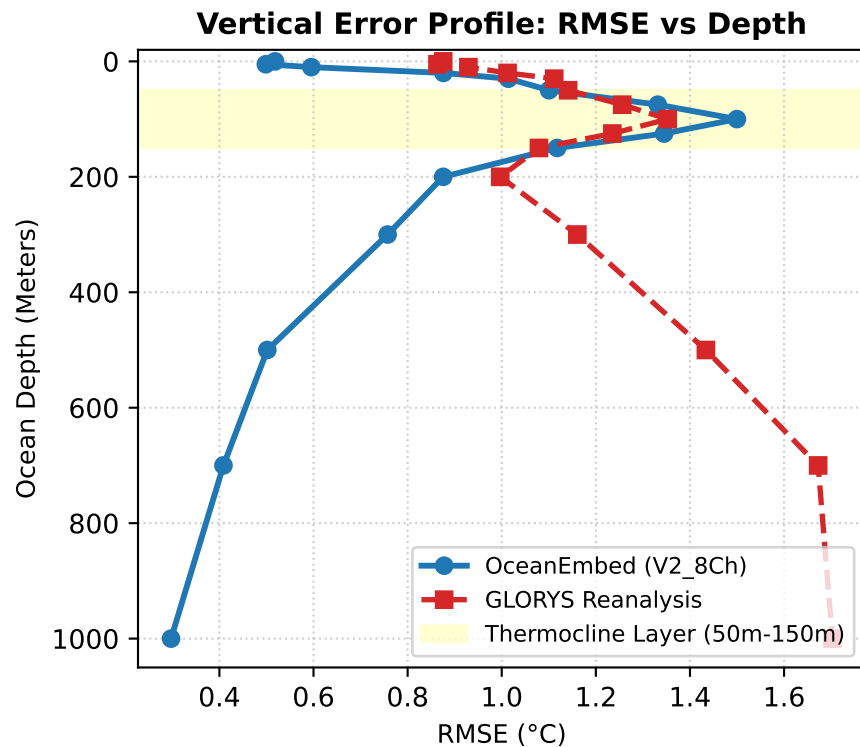


**Residual Error Frequency Distribution**



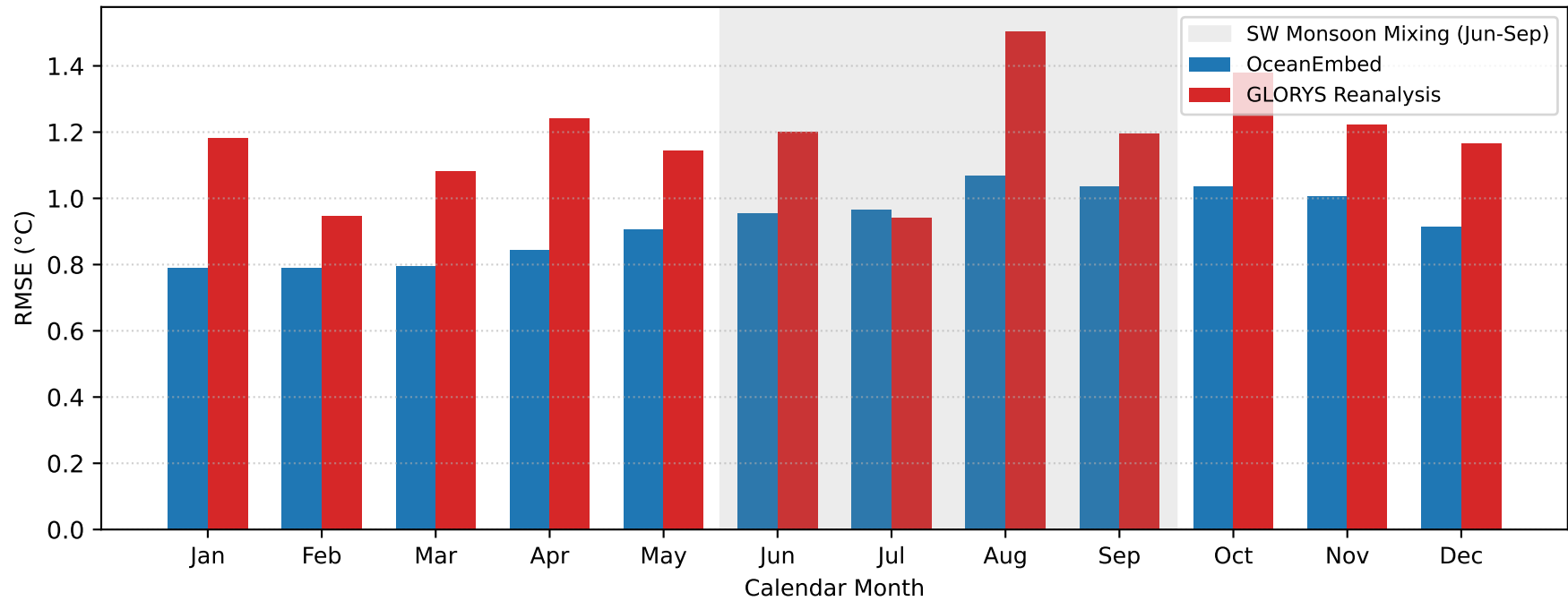
**Cumulative Absolute Error Distribution**



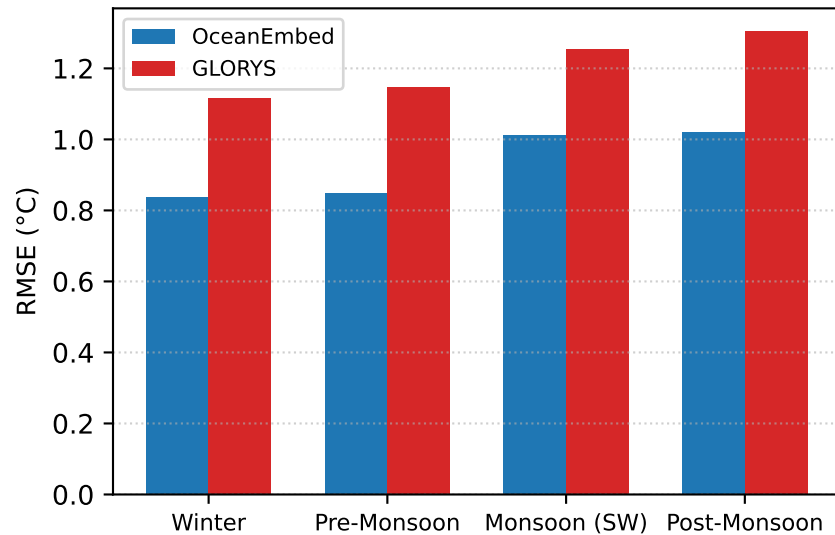


| Depth (m) | In-Situ Obs | OceanEmbed RMS | GLORYS RMSE | OceanEmbed MAE | GLORYS MAE | Dynamic Zone    |
|-----------|-------------|----------------|-------------|----------------|------------|-----------------|
| 0m        | 8,058       | 0.5184°C       | 0.8756°C    | 0.3317°C       | 0.3434°C   | Surface / Mixed |
| 5m        | 8,058       | 0.4987°C       | 0.8634°C    | 0.3161°C       | 0.3384°C   | Surface / Mixed |
| 10m       | 8,056       | 0.5954°C       | 0.9292°C    | 0.3391°C       | 0.3588°C   | Surface / Mixed |
| 20m       | 8,063       | 0.8761°C       | 1.0124°C    | 0.4792°C       | 0.4595°C   | Surface / Mixed |
| 30m       | 8,256       | 1.0141°C       | 1.1115°C    | 0.6194°C       | 0.5856°C   | Surface / Mixed |
| 50m       | 8,279       | 1.1006°C       | 1.1412°C    | 0.7472°C       | 0.7066°C   | Thermocline     |
| 75m       | 8,297       | 1.3316°C       | 1.2560°C    | 0.9795°C       | 0.8881°C   | Thermocline     |
| 100m      | 8,293       | 1.4997°C       | 1.3526°C    | 1.1621°C       | 1.0099°C   | Thermocline     |
| 125m      | 8,289       | 1.3451°C       | 1.2351°C    | 1.0614°C       | 0.9204°C   | Thermocline     |
| 150m      | 8,288       | 1.1177°C       | 1.0790°C    | 0.8666°C       | 0.7759°C   | Thermocline     |
| 200m      | 8,280       | 0.8756°C       | 0.9969°C    | 0.6315°C       | 0.6147°C   | Thermocline     |
| 300m      | 8,244       | 0.7575°C       | 1.1607°C    | 0.4679°C       | 0.5329°C   | Deep Ocean      |
| 500m      | 8,117       | 0.5018°C       | 1.4339°C    | 0.2718°C       | 0.4639°C   | Deep Ocean      |
| 700m      | 8,059       | 0.4086°C       | 1.6724°C    | 0.2411°C       | 0.5044°C   | Deep Ocean      |
| 1000m     | 6,371       | 0.2973°C       | 1.7027°C    | 0.2017°C       | 0.4940°C   | Deep Ocean      |

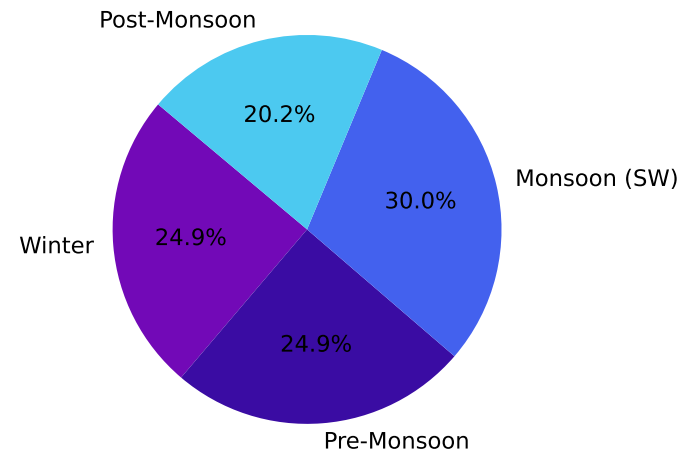
**Month-by-Month Predictive Stability Across Diverse Hydrographic Regimes**



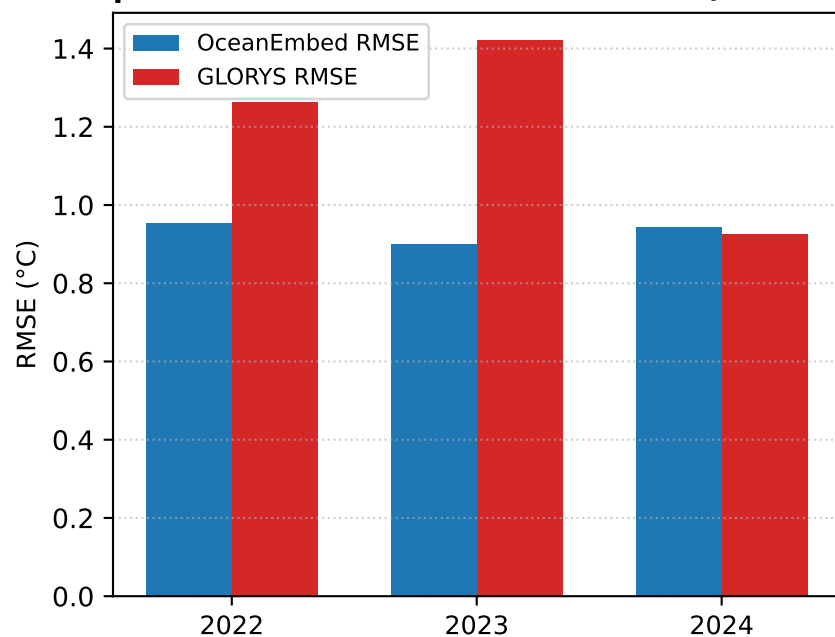
**Seasonal Regime Breakdown**



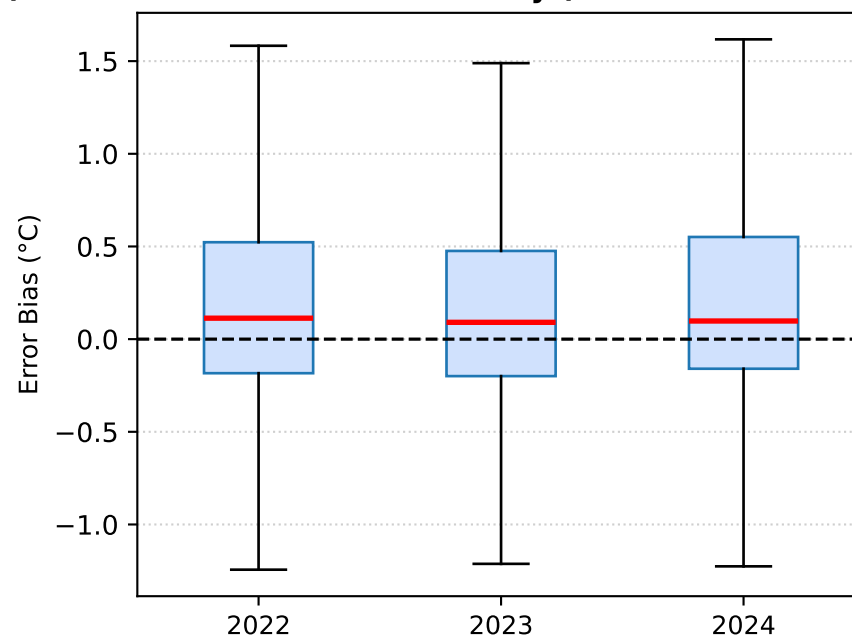
**In-Situ Validation Profile Density by Season**



**Temporal Generalization: Unseen Years (2022-2024)**



**Bias Distribution Stability (Zero Median Drift)**



| Evaluation Year | In-Situ ARGO PoDs | OceanEmbed RMSE | GLORYS RMSE | OceanEmbed MAE | GLORYS MAE | Early Stability Verified     |
|-----------------|-------------------|-----------------|-------------|----------------|------------|------------------------------|
| 2022            | 41,380            | 0.9525°C        | 1.2625°C    | 0.6023°C       | 0.7368°C   | Yes (Holdout Generalization) |
| 2023            | 36,134            | 0.9004°C        | 1.4203°C    | 0.5738°C       | 0.7120°C   | Yes (Holdout Generalization) |
| 2024            | 43,494            | 0.9426°C        | 0.9239°C    | 0.5914°C       | 0.7158°C   | Yes (Holdout Generalization) |